

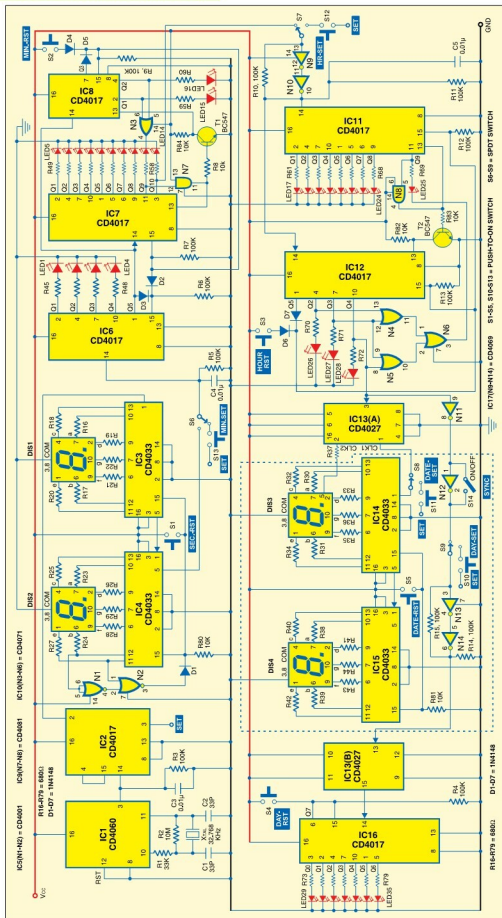


Fig. 4: Circuit for analogue clock with add-on calendar

ter and display circuit, except for a few changes like use of 3-input OR gate (made by three unused gates of IC10).

IC11 gets a PPH input from the previous stage. Before connecting the PPH input to clock input (pin 14) of IC11, a circuit built around N9 and N10 inverter gates (shown in Fig. 4) eliminates any contact-bounce effect while setting the hour time. Q1 through Q8 outputs of IC11 display 1 through 8 hours, respectively. Q9 output displays 9th hour and, at the same time, disables IC11 through the AND gate output and enables IC12 for counting by driving transistor T2, to light up 10th-, 11th- and 12th-hour LEDs at Q2, Q3 and Q4 outputs of IC12, respectively. Q5 output resets IC12 and IC11 and one-hour LED glows. Q4 output of IC12 provides a pulse every twelve hours, which is used as clock pulse for the date and day display circuits.

Switch S3 resets the hours display to '00' when pushed momentarily. You can set the required hours time by connecting SPDT switch S7 to switch S12 and press switch S12 until the desired hour LED glows.

Date display. This